```
.transition()
.duration(1000)
.attrTween("d", arcTween);

    function arcTween(a) {
       var i = d3.interpolate(this._current, a);
       this._current = i(0);
     return function(t) {
     return newArc(i(t));
     };
   }
```

Calls a tween on the d attribute

Uses the arc generator to tween the arc by calculating the shape of the arc explicitly

The result of the code in listing 5.2 is a pie chart that cleanly transitions the individual arcs or removes them when no data corresponds to the pie pieces. You'll see more of `attrTween` and `styleTween`, as well as a deeper investigation of easing and other transition properties, in later chapters.

We could label each pie piece `<path>` element, color it according to a measurement or category, or add interactivity. But rather than spend a chapter creating the greatest pie chart application you've ever seen, we'll move on to another kind of layout that's often used: the circle pack.

5.3 Pack layouts

Hierarchical data is amenable to an entire family of layouts. One of the most popular is circle packing, shown in figure 5.9. Each object is placed graphically inside the

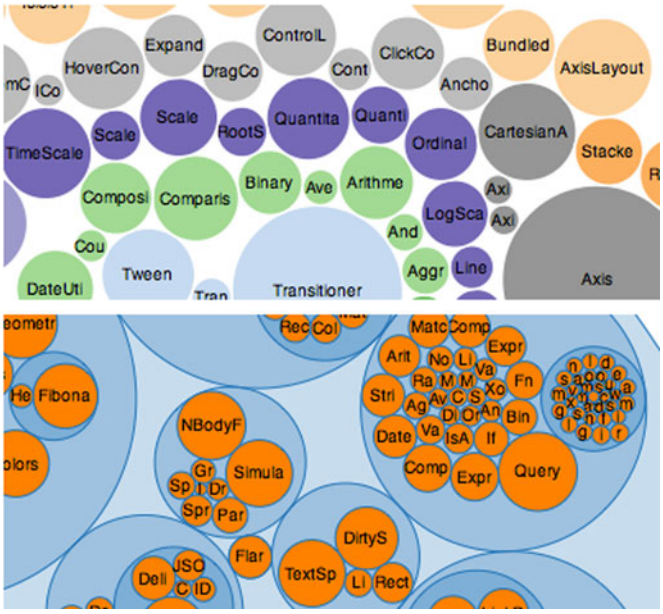


Figure 5.9 Pack layouts are useful for representing nested data. They can be flattened (top), or they can visually represent hierarchy (bottom). (Examples from Bostock, https://github.com/mbostock/d3/wiki/Pack-Layout.)